1 Solution — GIAM 8.1 Problem 4

1.1 Problem

Place the sets $\mathbb{N}$, $\mathbb{R}$, $\mathbb{Q}$, $\mathbb{Z}$, $\mathbb{Z} \times \mathbb{Z}$, $\mathbb{C}$, $\mathbb{N}_{2007}$, and $\emptyset$ somewhere on the Venn diagram from Problem 3.

The textbook notes that "there are no wrong answers — the point is to see what your intuition says at this point." Below we *do* place each set definitively (with one-line justifications), since the actual classifications are not difficult to argue, but all of them rest on the bijection / diagonal-argument machinery of Sections 8.2 onward, which is fair to defer.

1.2 Placement Diagram

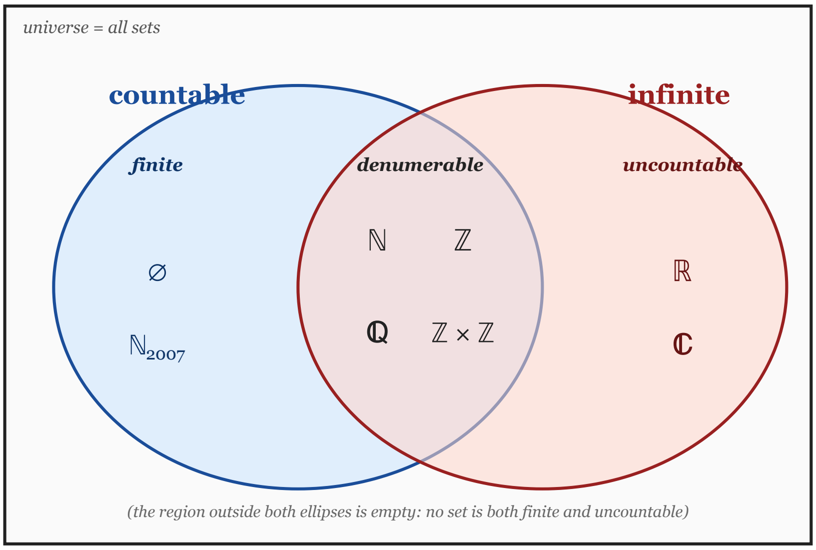

Figure 1.1: Placement of the eight sets onto the two-ellipse Venn diagram of Problem 3. Left ellipse = countable, right ellipse = infinite. Each set is dropped into one of the three non-empty atomic regions.

1.3 Justifications

Set	Region	One-line justification
$\emptyset$	finite	The empty set has 0 elements; it is the unique set of cardinality 0.
$\mathbb{N}_{2007}$	finite	By definition $\mathbb{N}_{2007} = \{1, 2, \ldots, 2007\}$ has exactly 2007 elements.
$\mathbb{N}$	denumerable	$\mathbb{N}$ is the canonical denumerable set — the identity map $\mathbb{N} \to \mathbb{N}$ is the witnessing bijection.
$\mathbb{Z}$	denumerable	The interleaving $0, 1, -1, 2, -2, 3, -3, \ldots$ gives a bijection $\mathbb{N} \to \mathbb{Z}$.
$\mathbb{Q}$	denumerable	Cantor's enumeration of $\mathbb{Q}$ as a list (skipping repeats) gives a bijection with $\mathbb{N}$. *(Proved in Section 8.2.)*
$\mathbb{Z} \times \mathbb{Z}$	denumerable	Cantor's snake / pairing function enumerates $\mathbb{Z} \times \mathbb{Z}$ as a single list, hence a bijection with $\mathbb{N}$. *(Proved in Section 8.2.)*
$\mathbb{R}$	uncountable	Cantor's diagonal argument shows no list of reals can be exhaustive. *(Proved in Section 8.2.)*
$\mathbb{C}$	uncountable	$\mathbb{R} \subseteq \mathbb{C}$, and any superset of an uncountable set is uncountable (else a sublist would enumerate $\mathbb{R}$).

Table 1.1

1.4 Region Summary

Reading off the three atomic regions of the diagram:

$$\text{finite}: \quad \{\emptyset,\ \mathbb{N}_{2007}\}$$

$$\text{denumerable}: \quad \{\mathbb{N},\ \mathbb{Z},\ \mathbb{Q},\ \mathbb{Z} \times \mathbb{Z}\}$$

$$\text{uncountable}: \quad \{\mathbb{R},\ \mathbb{C}\}$$

In the language of cardinalities:

$$|\emptyset| = 0, \qquad |\mathbb{N}_{2007}| = 2007,$$

$$|\mathbb{N}| = |\mathbb{Z}| = |\mathbb{Q}| = |\mathbb{Z} \times \mathbb{Z}| = \aleph_0,$$

$$|\mathbb{R}| = |\mathbb{C}| = \mathfrak{c} = 2^{\aleph_0}.$$

1.5 Remark — Surprises Worth Noting

Two placements above are likely *counterintuitive* on first encounter, even though the diagram makes them look unsurprising:

- $\mathbb{Q}$ *is denumerable.* The rationals are dense in $\mathbb{R}$ — between any two reals there are infinitely many rationals — yet $|\mathbb{Q}| = |\mathbb{N}|$. Density of a set in $\mathbb{R}$ has *no bearing* on cardinality.

- $\mathbb{Z} \times \mathbb{Z}$ *is denumerable.* Pairs of integers feel "two-dimensional" — like a grid — yet the grid can be enumerated by snaking through diagonals, putting it in bijection with the 1-dimensional $\mathbb{N}$. More generally, $A \times B$ is denumerable whenever A, B are denumerable; iterating gives $\mathbb{Z}^k$ denumerable for *any* finite k.

The reason the diagram makes them look unsurprising is precisely the point of the equivalence relation $\equiv$: it forgets every structural feature of a set (density, dimension, algebraic structure, ...) and remembers *only* the bijection class. From that lens, "rationals" and "natural numbers" really are the same object.

1.6 Remark — Why $\mathbb{C}$ Goes With $\mathbb{R}$, Not Above It

A naive guess is that $\mathbb{C}$, being "two-dimensional" over $\mathbb{R}$, ought to be *strictly larger* than $\mathbb{R}$ — analogous to how $\mathbb{R}$ is strictly larger than $\mathbb{N}$. But the bijection $\mathbb{R}^2 \to \mathbb{R}$ (e.g., interleaving digits of two decimals into one decimal) shows $|\mathbb{R}^2| = |\mathbb{R}|$, and hence $|\mathbb{C}| = |\mathbb{R}^2| = |\mathbb{R}| = \mathfrak{c}$. So $\mathbb{R}$ and $\mathbb{C}$ live in the *same* uncountable region — not stacked above one another. The next "level up" in cardinality would be $\mathcal{P}(\mathbb{R})$, with $|\mathcal{P}(\mathbb{R})| = 2^{\mathfrak{c}} > \mathfrak{c}$.

1.7 Remark — The Hierarchy of Sizes Visible Here

Among the eight sets, exactly three distinct cardinalities appear:

Cardinality	**Sets in this slot**
0	$\emptyset$
2007	$\mathbb{N}_{2007}$
$\aleph_0$	$\mathbb{N},\ \mathbb{Z},\ \mathbb{Q},\ \mathbb{Z}\times\mathbb{Z}$
$\mathfrak{c}$	$\mathbb{R},\ \mathbb{C}$

Table 1.2

So we see four cardinalities total $(0, 2007, \aleph_0, \mathfrak{c})$, but only three *region-classes* in the Venn diagram (finite, denumerable, uncountable). The Venn diagram collapses all finite cardinalities $(0, 1, 2, \dots)$ into one region, and *might* collapse multiple uncountable cardinalities into one region if larger uncountables were present (e.g., $\mathcal{P}(\mathbb{R})$) — but the eight sets here all land in $\mathfrak{c}$ or below.